CHAPTER 5

Activities & Lab Packet

Earth's Surface

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 5. Works with both the Standard and Easy Read versions of the textbook.

📘 Georgia Standard S6E5

WHAT'S INSIDE

1 The Crayon Rock Cycle

LAB

OBJECTIVE
Turn crayon shavings into all three rock types and act out the full rock cycle.

TIME
50 minutes

STANDARDS
S6E5.c

Materials (per group of 3–4)

- 3–4 old crayons in different colors, paper removed
- Plastic knives or pencil sharpeners (for shavings), squares of aluminum foil
- Books or a heavy block (pressure!), a warm-water bath (teacher station with hot tap water in a tray)
- Teacher-only station: hot plate or candle warmer for full melting (demo)
- Student data sheet (next page)

Setup

Crayons = rock. Shaving them = weathering. Each group makes its "rocks" in sequence, saving a small sample of each stage for the data sheet. The final melt is a teacher demo for safety.

Student Instructions

1. **Weather your rock:** shave the crayons into small bits (sediment!). Mix the colors.
2. **Make SEDIMENTARY rock:** pile the shavings in folded foil and press HARD under books for 2 minutes — no heat. Open it: layers pressed together. Sketch it and save a piece.
3. **Make METAMORPHIC rock:** wrap another pile in foil, dip in the warm-water bath 20–30 seconds, then squeeze firmly. Heat + pressure, but no melting. Open, sketch, save. Notice the colors swirl but stay separate.
4. **Make IGNEOUS rock (teacher demo):** watch as a foil boat of shavings is fully melted, then cooled. The colors blend completely into one new solid. Sketch the result.
5. Complete the data sheet, matching every step to the real rock cycle.

⚠️ SAFETY & TEACHER NOTES

Safety: Teacher handles all full melting; melted crayon is HOT wax. Warm-water bath should be hot-tap warm, not boiling. Blades supervised.
The key compare: Metamorphic vs. igneous look totally different — swirled-but-separate vs. fully blended. That's the "changed WITHOUT melting" idea made visible.
Cleanup: Foil contains everything. Cooled "igneous" crayons make great door prizes.

ANSWER KEY (data sheet)

1) Shaving = weathering; moving shavings to foil = erosion/transport; piling = deposition | 2) Sedimentary — pressure pressing layers (no heat) | 3) Metamorphic — heat AND pressure, no melting; bits deformed/swirled but still visible | 4) Igneous — fully melted then cooled; original pieces gone, one new solid | Conclusion: any rock can become any other rock given the right combination of weathering, pressure, heat, melting, and cooling.

The Crayon Rock Cycle — Data Sheet

Name: ____________________ Period: __________

Stage	Sketch	What we did (process) + what it models
Shavings (sediment)		
"Sedimentary" crayon		
"Metamorphic" crayon		
"Igneous" crayon		

1. Shaving the crayons modeled which real process? Moving the shavings? Piling them up?

2. What ingredient(s) made the sedimentary rock — heat, pressure, or both?

3. In the metamorphic crayon, could you still see the original colored bits? What does "changed WITHOUT melting" mean?

4. How was the igneous crayon different from the other two?

Conclusion: Could your igneous crayon be turned back into a sedimentary crayon? Describe the steps using rock-cycle words.

2 Shake, Break & Carry — Weathering & Erosion Stations

LAB

OBJECTIVE
Model weathering, erosion, and deposition at two hands-on stations and identify each process in action.

TIME
45–50 minutes

STANDARDS
S6E5.d, S6E5.e

Materials

- **Station A — Sugar-Cube Shaker:** 3 sugar cubes per group, a hard plastic jar with a lid, dark paper
- **Station B — Stream Table:** a paint-roller tray or baking pan filled with damp sand/soil at one end, a cup of water, a small pitcher, a few "houses" (cubes or game pieces)
- Student data sheet (next page)

Station A — Sugar-Cube Shaker (weathering)

1. Sketch a fresh sugar cube — sharp corners, flat faces.
2. Put 3 cubes in the jar, lid ON, and shake hard for 60 seconds (count!).
3. Pour onto dark paper. Sketch the cubes now and describe the "sand" that appeared. Shake 60 more seconds and compare.

Station B — Stream Table (erosion & deposition)

1. Build a sloped "hill" of damp sand at the high end of the tray. Place a "house" on the slope and one at the bottom.
2. Slowly pour water from the top in one steady spot. Watch the path the water cuts.
3. Find where sand was picked up (erosion) and where it dropped (deposition — look for the fan at the bottom). Sketch and label both.
4. Repeat with a faster pour. What changes? Now lay grass/paper strips ("plants") on a rebuilt slope and pour again — what do the plants do?

⚠️ SAFETY & TEACHER NOTES

Safety: Water + floors = slips; towels at Station B. No eating the sugar (lab rule!).
Vocabulary check: Station A is weathering ONLY (breaking in place); Station B adds erosion (moving) and deposition (dropping). Make students point to each.
Human connection (S6E5.e): The plant strips model ground cover — removing plants speeds erosion; this is why farmers plant cover crops and terrace hillsides.

ANSWER KEY (data sheet)

A: corners rounded, edges smoothed, sugar "sand" collects — like rocks tumbling in a river; more shaking = more wear | B: water cut a channel (erosion) and built a fan of dropped sand at the bottom (deposition); faster water moves more, bigger sediment; plants slowed the water and held sand in place | Conclusion: weathering breaks, erosion carries, deposition drops — and plants/human choices change how fast it happens.

Shake, Break & Carry — Data Sheet

Name: ______________________ Period: __________

STATION A — Sugar-Cube Shaker

Before shaking	After 60 sec	After 120 sec

1. What happened to the cubes' corners and edges? Where did the "sand" come from?

2. This station models WEATHERING. What real-world force does the shaking stand for?

STATION B — Stream Table

Sketch your stream table AFTER pouring — label EROSION (where sand left) and DEPOSITION (where it piled up)

3. What happened when you poured FASTER?

4. What did the "plants" do to erosion? What does that teach farmers and builders?

Conclusion: Use all three words — weathering, erosion, deposition — to tell the story of one grain of sand from your hill to the fan at the bottom.

3 Graham Cracker Plate Tectonics

LAB

OBJECTIVE	TIME	STANDARDS
Model the three plate movements and connect each to mountains, new crust, and earthquakes.	40 minutes	S6E5.a, S6E5.f

Materials (per group of 2–3)

- 1 paper plate spread thickly with frosting (or whipped topping) — this is the slowly-flowing **mantle**
- 4 graham cracker squares — these are **tectonic plates** (crust)
- 1 small cup of water (for dipping cracker edges), napkins
- Student data sheet (next page)

Setup

Frosting must be thick enough that crackers FLOAT and slide on it — just like plates riding the mantle. Yes, they may eat the lab at the end. Science has perks.

Student Instructions

1. **Float test:** set two crackers on the frosting and push gently — feel them glide? That's the crust riding the mantle. (Real speed: fingernail growth!)
2. **COLLIDE (push together):** dip the touching edges of two crackers in water for 2 seconds, set them on the frosting, and slowly push them together. Watch the soft edges crumple and rise — mountains! Sketch it.
3. **SEPARATE (pull apart):** place two dry crackers touching, then slowly pull them apart. What oozes up into the gap? That's magma rising at a mid-ocean ridge, making new crust. Sketch it.
4. **SLIDE PAST:** press two dry crackers edge-to-edge and slide them in opposite directions. Feel them catch… catch… then JERK — that jolt is an earthquake! Sketch it.
5. Complete the data sheet, matching each move to its real-world result.

⚠️ SAFETY & TEACHER NOTES

Allergy check: Confirm graham/frosting ingredients against class allergy lists; gluten-free crackers work fine.
Vocabulary anchor: As they work, require the sentence frame: "When plates ______, we get ______."
Ring of Fire tie-in: Project a map of earthquakes/volcanoes around the Pacific afterward — ask why the dots trace plate edges.

ANSWER KEY (data sheet)

1) They slide easily — crust floats on the soft, slow-flowing mantle | 2) Collide → edges crumple upward = mountains (and volcanoes where one dives under) | 3) Separate → frosting ("magma") rises into the gap = new crust, mid-ocean ridges/rift valleys | 4) Slide past → they stick then suddenly slip = earthquake | 5) Plate edges | Conclusion: slow plate movement over millions of years builds mountains, makes new sea floor, and triggers quakes — mostly at the boundaries (Ring of Fire).

Graham Cracker Plate Tectonics — Data Sheet

Name: ______________________________ Period: ______________

In this model: frosting = ______________________ crackers = ______________________

Plate movement	Sketch	What happened + real-world result
COLLIDE (push together)		
SEPARATE (pull apart)		
SLIDE PAST		

1. In the float test, why could the crackers move at all? What layer lets real plates drift?

____

2. In the slide-past test, the crackers stuck and then suddenly jerked. What is that sudden slip called in real life?

____

3. Where do most earthquakes and volcanoes happen — the middles of plates or the edges? What is the famous example circling the Pacific?

____

____

Conclusion: Real plates move only a few centimeters a year. Explain how something SO slow can build something as big as a mountain range.

____

____

4 Journey to the Center of the Earth

WORKSHEET

OBJECTIVE
Identify Earth's four layers and what changes with depth.

TIME
15–20 minutes

STANDARDS
S6E5.a

Name: ______________________________ Period: ______________

Part A. Label the layers from the SURFACE inward (1 = outermost, 4 = center):

____ Inner core ____ Outer core

____ Crust ____ Mantle

Part B. Which layer am I?

5. I'm thin, solid rock — you live on me: ______________________

6. I'm the THICKEST layer; my hot rock flows super slowly: ______________________

7. I'm liquid iron and nickel: ______________________

8. I'm the HOTTEST layer, but pressure squeezes me solid: ______________________

Part C. Complete the table — what happens to each as you go DEEPER?

Property	Goes UP or DOWN with depth?	Why? (one short reason)
Temperature		
Pressure		
Density		

Part D. Short answers:

9. The inner core is hotter than the outer core, yet the inner core is SOLID while the outer core is LIQUID. Explain how that's possible.

____

____

10. The textbook compares the crust to part of an apple. Which part, and why is it a good comparison?

____

ANSWER KEY

A: crust 1, mantle 2, outer core 3, inner core 4 | 5) crust 6) mantle 7) outer core 8) inner core | C: all three go UP — deeper = hotter, more weight pressing down, material packed tighter | 9) The enormous pressure at the center squeezes the metal solid even at the highest temperature | 10) The skin — extremely thin compared with everything underneath.

5 Rock Type Detective

WORKSHEET

OBJECTIVE
Classify rocks by how they form, and tell minerals from rocks.

TIME
20 minutes

STANDARDS
S6E5.b, S6E5.c

Name: ______________________________ Period: ______________

Part A. Write **I** (igneous), **S** (sedimentary), or **M** (metamorphic):

____ 1. Formed when magma or lava cools and hardens

____ 2. Layers of sediment pressed together

____ 3. Changed by heat + pressure WITHOUT melting

____ 4. Granite, obsidian, basalt

____ 5. Sandstone, limestone

____ 6. Marble (made from limestone), slate

____ 7. The name means "fire"

____ 8. The name means "changed form"

Part B. Mineral or rock? Write **MIN** or **ROCK**:

______ 9. One pure "ingredient" with a single recipe

______ 10. A mixture of one or more minerals

______ 11. Quartz, gold, salt

______ 12. Granite (quartz + feldspar + more)

Part C. Name the mineral ID test described:

13. Scratching it to see how tough it is: ______________________

14. Scraping it to see the color of its powder: ______________________

15. Checking how shiny it is (metallic, glassy, dull): ______________________

16. Seeing if it breaks along smooth, flat faces: ______________________

Part D. 17. A rock melts deep underground, then cools back into solid rock. What type is it now — no matter what it was before? Explain.

____

____

ANSWER KEY

1) I 2) S 3) M 4) I 5) S 6) M 7) I 8) M | 9) MIN 10) ROCK 11) MIN 12) ROCK | 13) hardness 14) streak 15) luster 16) cleavage | 17) Igneous — any rock that melts and re-cools becomes igneous; the rock cycle lets any type become any other.

6 Weathering, Erosion, or Deposition?

WORKSHEET

OBJECTIVE	TIME	STANDARDS
Sort surface-shaping events into weathering, erosion, and deposition — by nature AND by humans.	15–20 minutes	S6E5.d, S6E5.e

Name: ______________________________ Period: ______________

Part A. Write **W** (weathering), **E** (erosion), or **D** (deposition):

____ 1. Ice freezes in a crack and splits a boulder

____ 2. A muddy river carries soil toward the sea

____ 3. Silt settles and builds a river delta

____ 4. Acid rain slowly dissolves a statue

____ 5. A glacier drags rocks down a valley

____ 6. Waves pile sand into a beach

____ 7. Tree roots crack a sidewalk

____ 8. A landslide carries earth downhill (gravity!)

Part B. Name the four main AGENTS of erosion (the "movers"):

9. _____________ 10. _____________ 11. _____________ 12. _____________

Part C. Humans change the surface too! For each activity, does it SPEED UP or SLOW DOWN erosion?

Human activity	Speeds up or slows down erosion?
13. Clearing a forest for a parking lot	
14. Terracing a farmed hillside into steps	
15. Planting cover crops on bare fields	

Part D. 16. Use the memory trick to explain the difference between weathering and erosion. ("Wear down…" vs. "Exit…")

____

____

17. Which agent of erosion is the most powerful, and what famous canyon is the proof?

____

ANSWER KEY

1) W 2) E 3) D 4) W 5) E 6) D 7) W 8) E | 9–12) water, wind, ice (glaciers), gravity | 13) speeds up 14) slows down 15) slows down | 16) Weathering = wear down IN PLACE (breaking, no moving); erosion = EXIT, the pieces move away | 17) Moving water; the Grand Canyon.

7 Writing Prompts

WRITING

OBJECTIVE	TIME	STANDARDS
Use scientific vocabulary to explain concepts in writing.	15–30 minutes each	Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Elevator to the Core.** You ride a magic elevator from the surface to Earth's center. Describe each layer you pass through — and what happens to heat and pressure as you drop.
2. **Recipe vs. Ingredient.** Explain the difference between a mineral and a rock using the recipe/ingredient idea. Give one example of each.
3. **The Life of a Sand Grain.** Tell the story of one grain of sand: where it broke off (weathering), how it traveled (erosion), and where it finally landed (deposition).

Level 2 — Intermediate (3 paragraphs)

4. **Interview with a Rock.** A metamorphic rock is interviewed about its life story. It used to be sedimentary! Write the interview — it must describe both how it first formed and how heat and pressure changed it (without melting!).
5. **Mountains from a Crawl.** Tectonic plates move about as fast as your fingernails grow. Explain to a doubtful friend how something so slow can build the Himalayas, cause earthquakes, and create new sea floor.
6. **Seashells on the Summit.** Hikers find fossils of sea creatures near a mountaintop. Write the scientific story of how that's possible, using fossils, sedimentary rock, and plate tectonics as evidence.

Level 3 — Advanced (essay)

7. **People vs. Plates.** Both nature and humans reshape Earth's surface. Compare them: which changes are faster? Bigger? Easier to control? Argue whether humans should be called "a force of nature," using examples from mining, farming, and building.
8. **Read the Layers.** Scientists "read" rock layers and the fossils in them like pages of a book. Explain how layer depth tells age, what fossils reveal about past climates and surfaces, and why sedimentary rock is the best "library." Then describe one discovery this method made possible.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE
Practice scientific thinking through verbal reasoning.

USE
Warm-ups, exit tickets, Socratic seminar

STANDARDS
Cross-cutting

📘 **TEACHER NOTES**

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. No one has ever drilled past the crust. How do scientists know what's deeper?
2. Is ice a mineral? Is water? Defend your answer with the definition.
3. A gravestone from 1820 is hard to read; one from 2010 is crisp. What process is responsible?
4. Why doesn't Georgia get big earthquakes like California does?
5. You find a perfectly round, smooth stone in a creek. Tell its backstory in one sentence.

Deeper Discussions (5–15 min each)

6. The rock cycle has no beginning and no end. Is a rock ever really "finished"? What does that mean for the granite in a kitchen countertop?
7. The Atlantic Ocean gets a few centimeters wider every year as plates separate. What will maps look like in 100 million years? Can we know?
8. Cities keep being rebuilt in earthquake zones along the Ring of Fire. Why do people live there anyway — and what could make those cities safer?
9. Topsoil takes centuries to form but can wash away in one storm season. Who should be responsible for protecting it — farmers, governments, or everyone? Why?
10. Fossil evidence shows Georgia's coast was once underwater and polar regions were once warm. What does that suggest about how "permanent" today's map and climate are?

9 Earth's Surface Museum Exhibit

GROUP PROJECT

OBJECTIVE	TIME	STANDARDS
Build and present a museum-style exhibit explaining one process that shapes Earth's surface.	2–3 class periods + gallery walk	S6E5 (a–h)

Project Description

The classroom becomes the "Museum of the Changing Earth." Teams of 3–4 each draw an exhibit topic: **Earth's Layers • The Rock Cycle • Weathering & Erosion • Plate Tectonics • Volcanoes & Earthquakes • Fossils Tell a Story • The Soil Profile**. Each team builds a tabletop exhibit and acts as museum guides during a class gallery walk.

Exhibit Requirements

- A **3-D model or large labeled diagram** (clay layers, foam plates, shoebox soil profile, paper-mâché volcano…)
- A museum **placard**: exhibit title + a 3–4 sentence explanation in your own words
- At least **5 vocabulary words** used correctly on labels
- One **"Did You Know?"** fact card (e.g., the inner core is nearly as hot as the Sun's surface!)
- A 90-second guided tour speech — every member speaks

Gallery Walk Day

Half the teams guide while half tour, then swap. Visitors complete a "museum passport" — one new fact learned at every exhibit. Finish with a class vote: Best Model, Best Tour Guides, Best Did-You-Know.

📘 TEACHER NOTES

Differentiation: Offer template placards and a model menu for teams that need structure; advanced teams must add a "what if?" panel (e.g., what if plates stopped moving?).
Materials tip: Recycled boxes, clay, and yarn cover almost every topic. The soil-profile team can use real layers in a clear jar.
Grading rubric (suggested): Scientific accuracy (40%), required elements (25%), creativity (20%), presentation (15%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE	TIME	STANDARDS
Review chapter vocabulary in a low-stress game format.	20–30 minutes	S6E5 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

crust	igneous rock	weathering	volcano
mantle	sedimentary rock	erosion	fossil
outer core	metamorphic rock	deposition	soil
inner core	rock cycle	plate tectonics	organic matter
mineral	magma	tectonic plate	bedrock
rock	lava	earthquake	topsoil

		FREE ⛰️		

📘 **TEACHER NOTES**

Read definitions straight from the Vocabulary Reference sheet (Section 5) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.